WPE Layout Studio

Give it the floor and the line rate. It puts the machines, robots, fences and walkway in place and checks every clearance rule for you.

The problem

The brief asks for two welding stations in a fixed floor area, with strict clearances between robots, machines, fences, trolleys and people. Doing that by hand means hours of CAD edits, and one moved machine breaks a rule somewhere else.

What we built

A website. You type the hall size and the jobs-per-hour target. It finds how many robots you need, draws the layout, and marks every rule pass or fail. Drag anything and it re-checks instantly.

14.4 m	8 / 8	30–50	0
material travel, Station 1	clearance rules passing	layouts tried per run	installs — it is a web page

How to show it (90 seconds)

1. Open the site. Say: nothing is installed, it runs in the browser.
2. Station 1 at 60 jobs/hour: 3 robots, 2 machines, 8 trolleys. It already looks right — 14.4 m travel, all 8 rules pass.
3. Switch to “Two stations”. A second line appears with a shared walkway: 8 robots, still 8/8.
4. Drag a robot onto a machine. Rules drop to 6/8 on screen. Press Ctrl+Z and it is compliant again.
5. Finish on the export buttons: DXF for CAD, PNG for the slide, report for the review.

Before you present

1. Open the link once and leave the tab open. First load is instant, there is nothing to install.
2. Start on Station 1 with the hall at 24 × 13 m. Do not change the numbers unless someone asks.
3. If the Wi-Fi drops, open index.html from the project folder — the whole site works offline.

What to remember

You do not need the screen for this page.

Say this if you only get one sentence

> **SAY** We turned the layout rules into code, so the plan is checked while it is being drawn — not by eye afterwards.

The rules it checks

800 mm between robot and machine	600 mm between equipment and the safety fence
700 mm between two robots	900 mm clear space in front of every trolley
1200 mm walkway, one connected loop	100 mm between trolleys
No overlapping equipment	Trolleys stay on the hall perimeter

If someone asks

Is this AI?	No. It is rule-based search: it tries 30–50 layouts and keeps the one that passes every rule with the shortest travel. Easy to explain and easy to trust.
Does it read our drawing?	It reads DXF (any CAD can export that). The DWG itself needs one export step first.
What if nothing fits?	It tells you the size you actually need, instead of drawing a wrong layout.
Can we change the numbers?	Yes — every clearance and the number of robots are editable on screen.
Why trust the result?	The rules are re-measured from the drawing every time, and the checks are covered by automated tests.

Links — open it now, it works on any laptop

Live site: mr-snake-mr.github.io/wpe-layout-studio · Code: github.com/mr-snake-mr/wpe-layout-studio
No internet? Open **WPE-Layout/index.html** from the project folder.